Diogo Poeta

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PhD physicist with research experience in experimental photonics, optical instrumentation, and computational simulations.

RESEARCH EXPERIENCE

Research Assistant | INL, Braga Sep 2022 - Apr 2026

International Iberian Nanotechnology Laboratory | EU LEARN project

- Operated a high-power femtosecond Yb laser system with tunable UV-VIS pulses at 315-540 nm; investigated third-harmonic generation in dielectric nanomembranes for VUV generation.
- Acquired and analyzed experimental data using Python; used LabVIEW and National Instruments data acquisition hardware for instrument control, and integrated basic electronics into experimental setups.
- Contributed to the development of selective VOC photoionization sensing for indoor air monitoring in schools within a multidisciplinary EU project.
- Operated gas-handling systems with pressure and flow control, mass flow controllers and rotameters, including Aalborg equipment.
- Designed and simulated dielectric nanostructures using Lumerical FDTD and MEEP; investigated optical anapoles, field distributions and energy velocity to study nonlinear optical response.
- Fabricated dielectric thin films and characterized them using ellipsometry, UV–Vis–NIR spectrophotometry, XRD, XPS and AFM.
- Managed the institute's COMSOL Multiphysics licensing.
- Onboarded and trained new researchers.

Research Intern | INL, Braga Nov 2020 - Dec 2021

- Designed transition-metal dichalcogenide (TMDC) metasurfaces and plasmonic nanostructures using Lumerical FDTD to investigate optical transmission and field enhancement.
- Performed Raman and photoluminescence spectroscopy, spectroscopic ellipsometry and 2D-material transfer.

Invited Researcher in Proteomics | University of Beira Interior Feb 2017 - Jun 2018

- Conducted work on the biosynthesis, recovery and functional characterization of therapeutic proteins; routinely operated GC–MS for sample measurements.

EDUCATION

PhD in Physics | Bielefeld University, Germany 2023 - 2026

MSc in Micro and Nano Technology | University of Minho, Portugal 2019 - 2022

BSc in Bioengineering | University of Beira Interior, Portugal 2016 - 2019

PUBLICATIONS

- Vilarinho, R., Mahmoodnia, H., **Poeta, D**., et al. "Volatile organic compound photoionization using tunable quasi-monochromatic VUV down to 120 nm." *Optics Express* 33(12), 26342–26355, Jun 2025. DOI 10.1364/OE.564577
- Vilarinho, R., Mahmoodnia, H., **Poeta, D**., et al. "Real-time molecular-specific BTEX detection at TLV-TWA concentration levels." *Environmental Technology and Innovation* 42, 104895, 2026. DOI 10.1016/j.eti.2026.104895
- **Poeta, D**., Vilarinho, R., et al. "Optical anapoles excited by UV-A illumination." *Optics Letters* 49(24), 7198–7201, Dec 2024. DOI 10.1364/OL.536925
- **Poeta, D.,** Maibohm, C., et al. "Energy velocity as a measure for the nonlinear response of TiO_2 in the ultraviolet." *Optical Materials Express* 16(10), 3342, 2026. DOI 10.1364/OME.611814

TECHNICAL SKILLS

Optical Instrumentation: Femtosecond Yb lasers; free-space alignment; tunable UV-VIS generation; third-harmonic generation in dielectric nanomembranes.

Gas handling and VOC sensing: Photoionization detection; gas delivery; pressure and flow control; mass flow controllers; rotameters; saturated vapor pressure principles; routine GC–MS operation in earlier proteomics research.

Instrumentation and Data Analysis: Python; AutoCAD; LabVIEW; National Instruments data-acquisition hardware and instrument control; basic electronics integration; electronic component soldering, including SMD components under a microscope.

Computational Photonics: Lumerical FDTD; MEEP; nanostructure and metasurface design; electromagnetic field calculations, multipole decomposition and energy-velocity analysis.

Micro-Nano Fabrication: Multi-target sputtering; optical lithography; dicing; wet processing; contact profilometry; atomic layer deposition (ALD).

Thin-film Characterization: Ellipsometry; UV–Vis–NIR; X-ray diffraction (XRD); X-ray photoelectron spectroscopy (XPS); atomic force microscopy (AFM); Raman and photoluminescence spectroscopy.

PRESENTATIONS

Apr 2025 | Poster | SPIE Optics + Optoelectronics, Metamaterials XV, Prague. TiO_2/SiO_2 metasurfaces for enhancement of THG with pump wavelengths in the UV-A.

Apr 2024 | Talk | 3MT, INL Annual Research Symposium, Braga. Optical anapoles excited by UV-A illumination.

Sep 2021 | Talk | Conferencia Española de Nanofotónica, virtual. Extraordinary transparency in continuous metallic thin film based on transition-metal-dichalcogenide metasurfaces.

LANGUAGES

Portuguese: native | English: professional | Spanish: professional